

Problem 1.1

```
a = [2; 5; 8]
```

```
a = 3×1
     2
     5
     8
```

```
b = [3; 13; 8]
```

```
b = 3×1
     3
    13
     8
```

```
A = a+b
```

```
A = 3×1
     5
    18
    16
```

```
% B = a*b error message
% invalid dimensions for matrix multiplication
C = a.*b
```

```
C = 3×1
     6
    65
    64
```

Problem 1.2

```
clear
a = [1 2 -3; 2 1 2; 4 -2 1]
```

```
a = 3×3
     1     2    -3
     2     1     2
     4    -2     1
```

```
b = [3; 13; 8]
```

```
b = 3×1
     3
    13
     8
```

```
A = a+b
```

```
A = 3×3
```

4	5	0
15	14	15
12	6	9

```
B = a*b
```

```
B = 3x1
      5
     35
    -6
```

```
C = a.*b
```

```
C = 3x3
      3      6     -9
     26     13     26
     32    -16      8
```

Problem 2a

```
t = 0:1:50
```

```
t = 1x51
      0      1      2      3      4      5      6      7      8      9     10     11     12
```

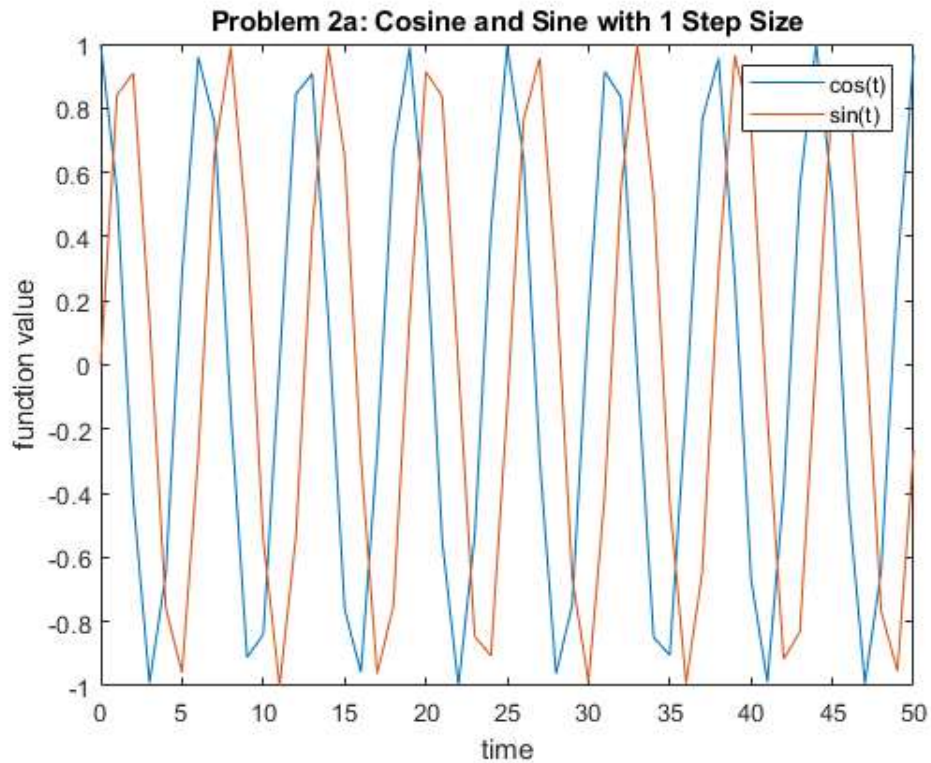
```
y1 = cos(t)
```

```
y1 = 1x51
      1.0000      0.5403     -0.4161     -0.9900     -0.6536      0.2837      0.9602      0.7539
```

```
y2 = sin(t)
```

```
y2 = 1x51
      0      0.8415      0.9093      0.1411     -0.7568     -0.9589     -0.2794      0.6570
```

```
plot(t, y1)
hold on;
plot(t, y2)
title('Problem 2a: Cosine and Sine with 1 Step Size')
xlabel('time')
ylabel('function value')
legend('cos(t)', 'sin(t)')
hold off;
```



Problem 2b

```
t = 0:0.01:50
```

```
t = 1×5001
    0    0.0100    0.0200    0.0300    0.0400    0.0500    0.0600    0.0700
```

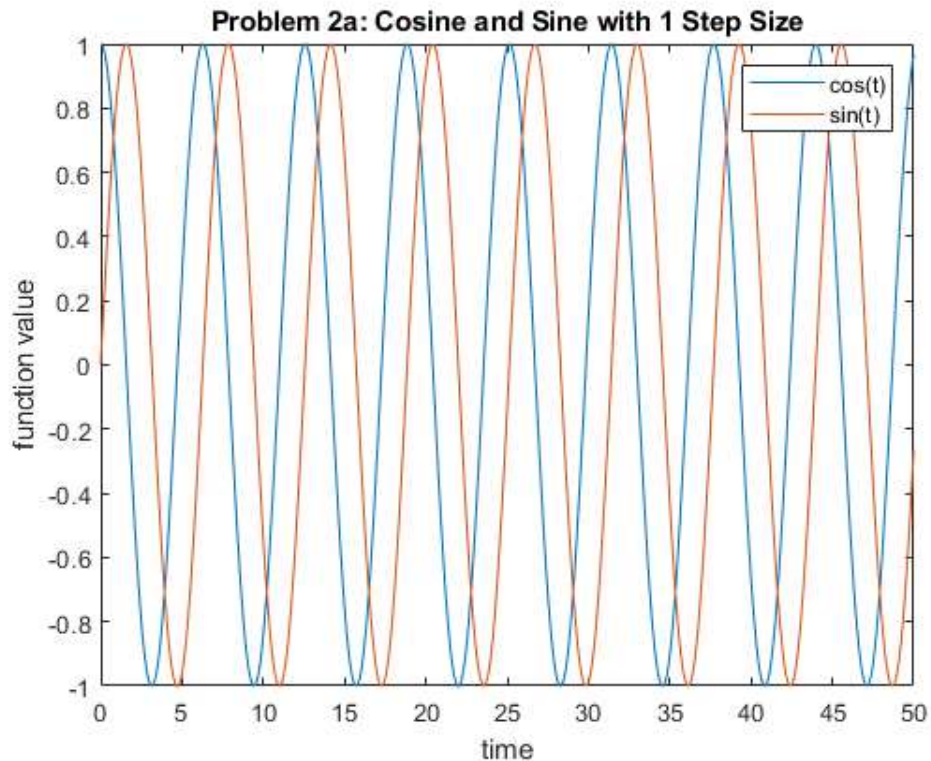
```
y1 = cos(t)
```

```
y1 = 1×5001
    1.0000    1.0000    0.9998    0.9996    0.9992    0.9988    0.9982    0.9976
```

```
y2 = sin(t)
```

```
y2 = 1×5001
    0    0.0100    0.0200    0.0300    0.0400    0.0500    0.0600    0.0699
```

```
plot(t, y1)
hold on;
plot(t, y2)
title('Problem 2a: Cosine and Sine with 1 Step Size');
xlabel('time');
ylabel('function value');
legend('cos(t)', 'sin(t)')
hold off;
```



The plot with the smaller step size results in a smoother looking graph.

Problem 3

```
clear
```

```
prompt = "Let's start with the first equation. Enter the coefficient for x: ";  
a1 = input(prompt)
```

```
a1 = 2
```

```
prompt = "Enter the coefficient for y: ";  
b1 = input(prompt)
```

```
b1 = 3
```

```
prompt = "Enter the coefficient for z: ";  
c1 = input(prompt)
```

```
c1 = 1
```

```
prompt = "Enter the coefficient for d: ";  
d1 = input(prompt)
```

```
d1 = 3
```

```
prompt = "Now the second equation. Enter the coefficient for x: ";  
a2 = input(prompt)
```

```
a2 = 1
```

```
prompt = "Enter the coefficient for y: ";
```

```
b2 = input(prompt)
```

```
b2 = 3
```

```
prompt = "Enter the coefficient for z: ";  
c2 = input(prompt)
```

```
c2 = -1
```

```
prompt = "Enter the coefficient for d: ";  
d2 = input(prompt)
```

```
d2 = 6
```

```
prompt = "Now the third equation. Enter the coefficient for x: ";  
a3 = input(prompt)
```

```
a3 = 2
```

```
prompt = "Enter the coefficient for y: ";  
b3 = input(prompt)
```

```
b3 = 2
```

```
prompt = "Enter the coefficient for z: ";  
c3 = input(prompt)
```

```
c3 = 0
```

```
prompt = "Enter the coefficient for d: ";  
d3 = input(prompt)
```

```
d3 = 7
```

```
A = [ a1 b1 c1 ; a2 b2 c2; a3 b3 c3 ]
```

```
A = 3×3
```

```
    2    3    1  
    1    3   -1  
    2    2    0
```

```
b = [ d1; d2; d3 ]
```

```
b = 3×1
```

```
    3  
    6  
    7
```

```
t = inv(A) * b
```

```
t = 3×1
```

```
    4.0000  
   -0.5000  
   -3.5000
```

Problem 4

```
clear
prompt = "Enter the value for mu: ";
mu = input(prompt)
```

```
mu = 255
```

```
prompt = "Enter the value for x: ";
x = input(prompt)
```

```
x = 1×101
      0      0.0100      0.0200      0.0300      0.0400      0.0500      0.0600      0.0700
```

```
y = (log(1 + (mu .* abs(x))) ./ log(1 + mu)) .* sign(x)
```

```
y = 1×101
      0      0.2285      0.3261      0.3891      0.4357      0.4727      0.5034      0.5290
```

```
plot(x, y)
title('Input-Output Characteristic of a Mu-Law Compressor');
xlabel('Input Signal');
ylabel('Compressed Output');
```

